Name: ___________________________

Date: ___________________________

Mr. Rawson • Y9 MYP Physics

1.8 – The Wave Equation

Unit 1: Oscillations & Waves

Learning intentions

- a) I can define **wave speed**, **frequency** and **wavelength**.
- b) I can use the equation $v = f \times \lambda$ to calculate wave speed.
- c) I can rearrange the equation to find frequency or wavelength.
- d) I can describe a method to measure the speed of a wave in a ripple tank or using echoes.

Theory

A **wave** transfers energy without transferring matter. The **wave speed** (v) is how fast the wave profile moves through a medium, measured in metres per second (m/s). The **frequency** (f) is the number of complete waves passing a point each second, measured in hertz (Hz). The **wavelength** (λ) is the distance between two corresponding points on adjacent waves, such as crest to crest, measured in metres (m).

The relationship between these three quantities is known as the **wave equation**: $v = f \times \lambda$ This means wave speed equals frequency multiplied by wavelength. If you know any two of these values, you can calculate the third by rearranging the formula: $f = v/\lambda$ $\lambda = v/f$

Worked Example 1 (Water Wave): A water wave has a frequency of 2 Hz and a wavelength of 1.5 m. What is its speed? Step 1: Identify known values: $f = 2$ Hz, $\lambda = 1.5$ m. Step 2: Choose the equation: $v = f \times \lambda$. Step 3: Substitute: $v = 2 \times 1.5$. Step 4: Calculate: $v = 3$ m/s.

Worked Example 2 (Sound Wave): A sound wave travels at 330 m/s with a frequency of 660 Hz. What is its wavelength? Step 1: Identify known values: $v = 330$ m/s, $f = 660$ Hz. Step 2: Rearrange equation: $\lambda = v/f$. Step 3: Substitute: $\lambda = 330/660$. Step 4: Calculate: $\lambda = 0.5$ m.

To measure wave speed practically, you can use a ripple tank or an echo method. In a **ripple tank**, you generate waves and measure the wavelength (λ) using a ruler on the screen. You then count the number of waves passing a point in one second to find the frequency (f). Multiply these values to get speed. Alternatively, for sound, you can stand a known distance (d) from a wall, clap once, and time the echo (t). The total distance travelled is $2d$, so speed $v = 2d/t$.

Complete the sentence: Wave speed is calculated by multiplying frequency by ___________ .

Word bank: wavelength · hertz · amplitude · period · crest

Activity

Measuring Wave Speed in a Ripple Tank

You will need: A ripple tank (or simulation), a ruler, a stopwatch, and a light source.

- Set up the ripple tank so that waves are generated at a steady frequency.
- Use the ruler to measure the distance between five consecutive crests on the projected screen. Divide this total distance by 4 to find the average **wavelength** (λ) in metres.
- Place your eye at a fixed point near the edge of the tank. Use the stopwatch to count how many waves pass that point in exactly 10 seconds. Divide the count by 10 to find the **frequency** (f) in Hz.
- Record your data in the table below.

Measurement	Value	Unit
Distance of 5 crests	____________	m
Wavelength (λ)	____________	m
Waves in 10 seconds	____________	count
Frequency (f)	____________	Hz
Calculated Speed (v)	____________	m/s

Questions

1. Define the term **wavelength** in your own words.

2. A wave has a frequency of 50 Hz and a wavelength of 0.2 m. Calculate its speed. Show your working.

3. Rearrange the equation $v = f \times \lambda$ to make frequency (f) the subject.

4. A sound wave travels through air at 340 m/s. If the frequency is 170 Hz, what is the wavelength? Show your working.

5. Explain why measuring the distance of five crests and dividing by four gives a more accurate wavelength than measuring just one crest-to-crest distance.

Extension

Research how the speed of sound changes in different states of matter (solids, liquids, gases). Explain why sound travels fastest in solids using the concept of particle spacing and energy transfer.